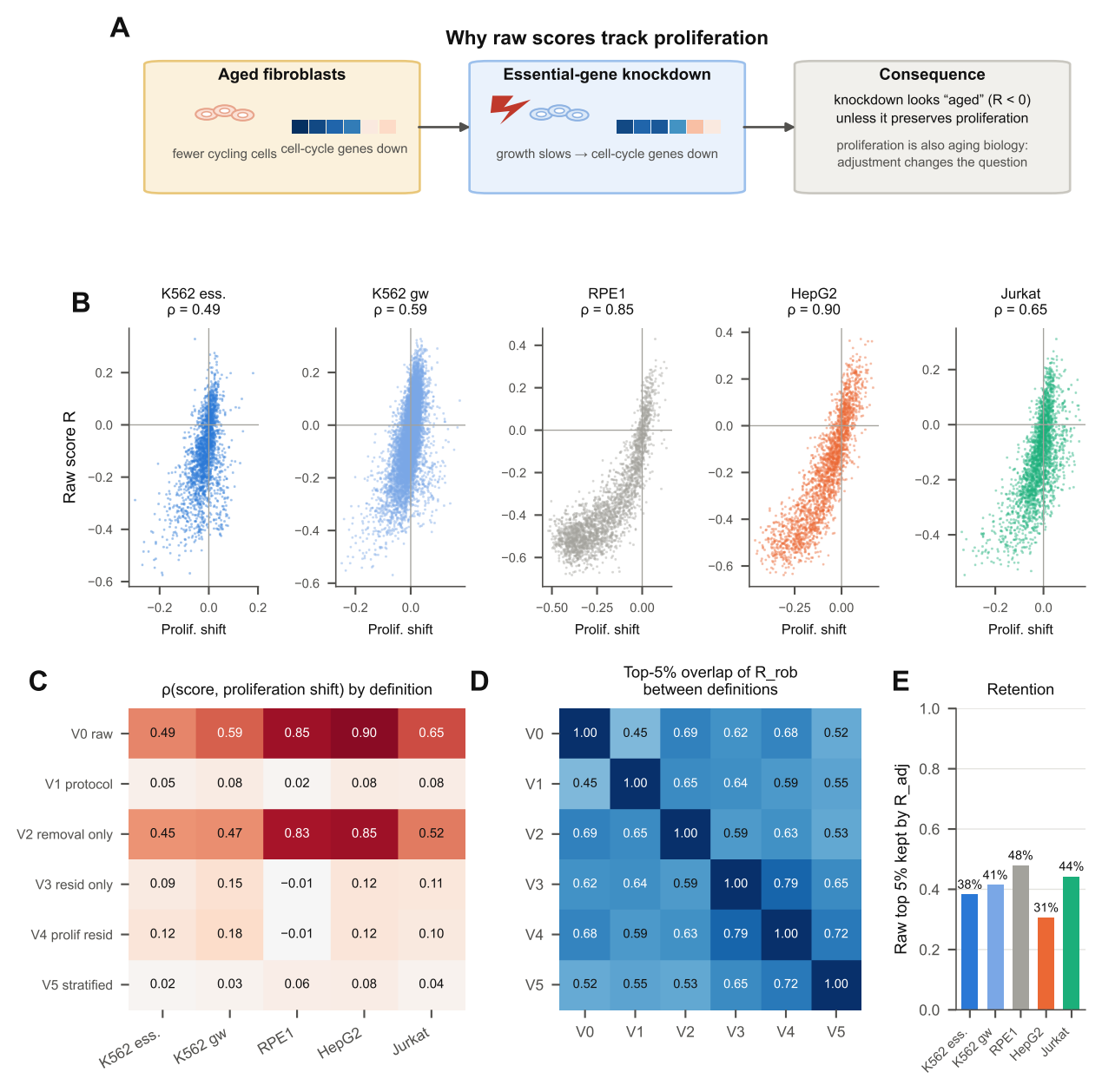

A
Why raw scores track proliferation
Aged fibroblasts
fewer cycling cells
cell-cycle genes down
Essential-gene knockdown
growth slows → cell-cycle genes down
Consequence
knockdown looks "aged" (R < 0)
unless it preserves proliferation
proliferation is also aging biology:
adjustment changes the question
B
K562 ess.
ρ = 0.49
K562 gw
ρ = 0.59
RPE1
ρ = 0.85
HepG2
ρ = 0.90
Jurkat
ρ = 0.65
Raw score R
Prolif. shift
C
ρ(score, proliferation shift) by definition
V0 raw 0.49 0.59 0.85 0.90 0.65
V1 protocol 0.05 0.08 0.02 0.08 0.08
V2 removal only 0.45 0.47 0.83 0.85 0.52
V3 resid only 0.09 0.15 −0.01 0.12 0.11
V4 prolif resid 0.12 0.18 −0.01 0.12 0.10
V5 stratified 0.02 0.03 0.06 0.08 0.04
K562 ess.
K562 gw
RPE1
HepG2
Jurkat
D
Top-5% overlap of R_rob
between definitions
V0 1.00 0.45 0.69 0.62 0.68 0.52
V1 0.45 1.00 0.65 0.64 0.59 0.55
V2 0.69 0.65 1.00 0.59 0.63 0.53
V3 0.62 0.64 0.59 1.00 0.79 0.65
V4 0.68 0.59 0.63 0.79 1.00 0.72
V5 0.52 0.55 0.53 0.65 0.72 1.00
E
Retention
Raw top 5% kept by R_adj
38%
41%
48%
31%
44%
K562 ess.
K562 gw
RPE1
HepG2
Jurkat